



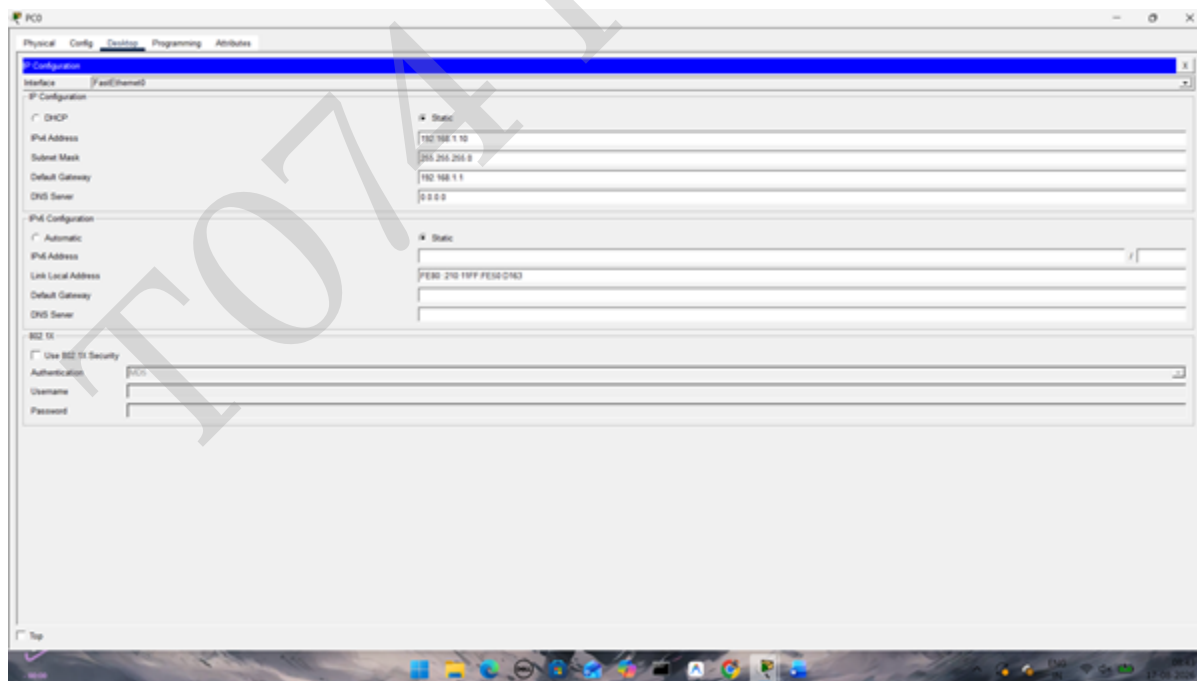
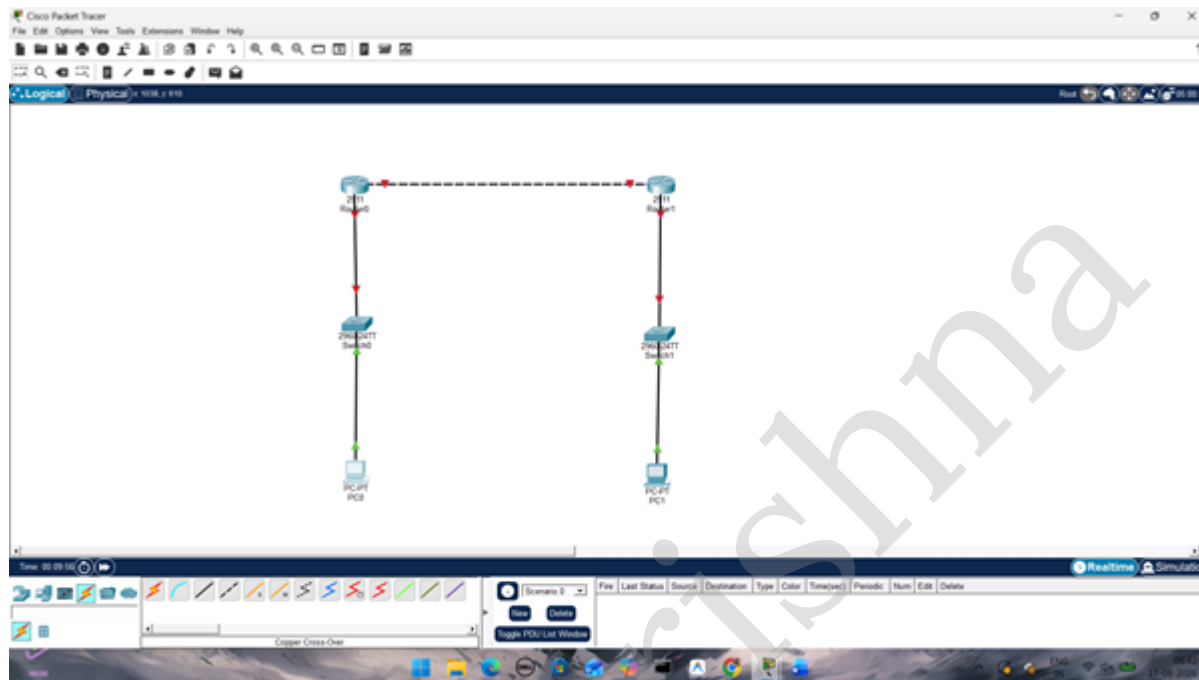
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DR. S. RADHAKRISHNAN MARG, ANDHERI(EAST), MUMBAI - 400069

PRACTICAL NO 6

AIM: Implement the Diffie-Hellman key exchange algorithm to securely exchange keys between two entities over an insecure network.



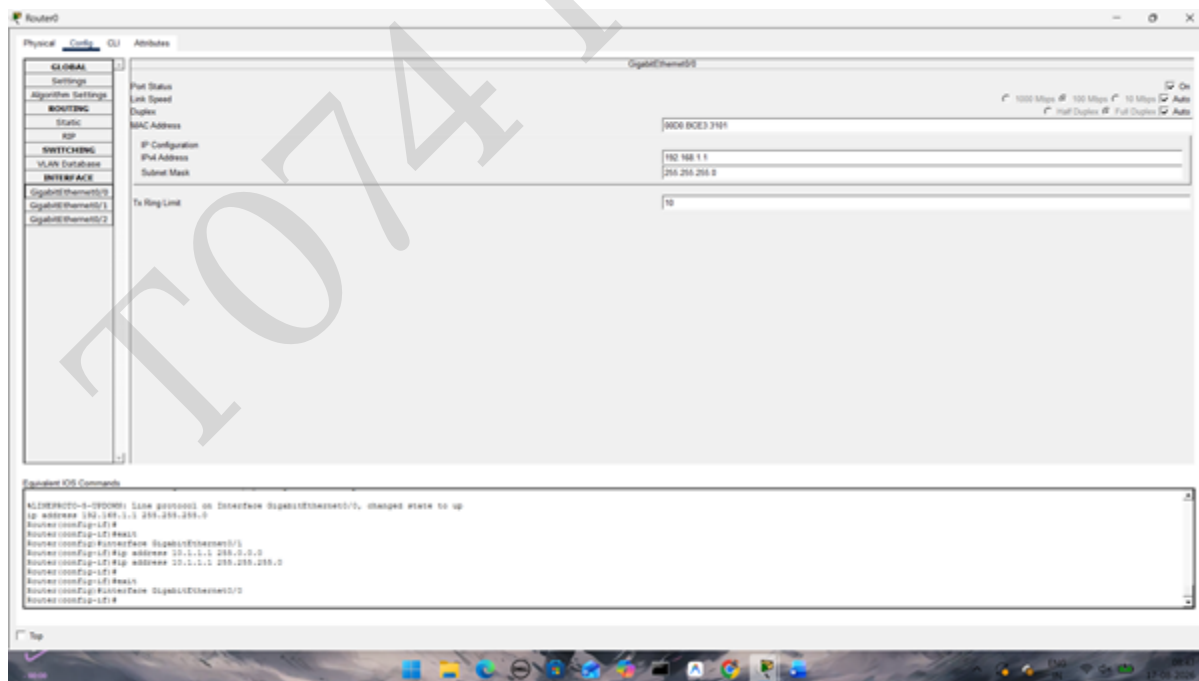
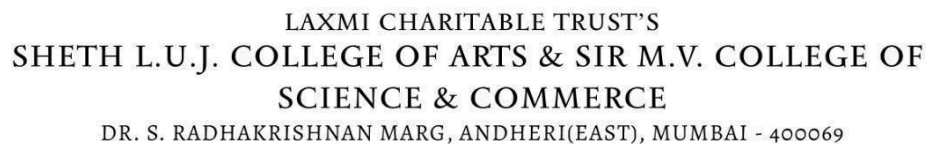
1. Network Topology & Devices



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CYBER SECURITY



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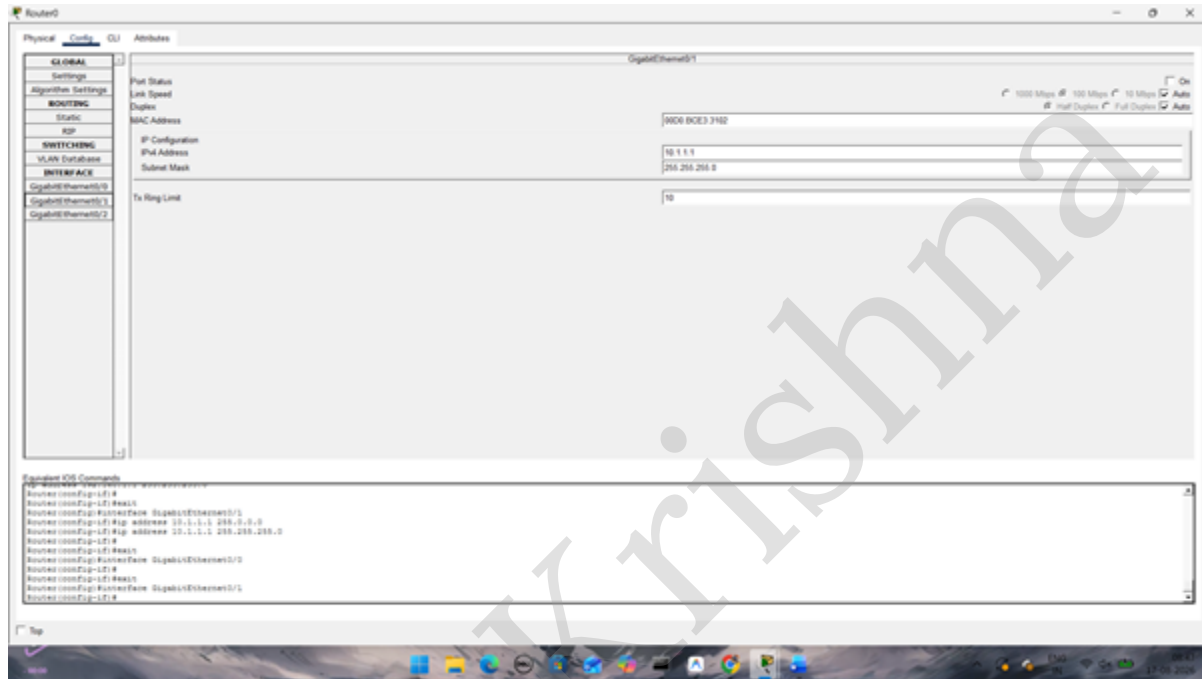


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IP Address Configuration

- **Site 1 (Left):**
 - **Router0:** LAN (Gig0/0) = 192.168.1.1 | WAN (Gig0/1) = 10.1.1.1
 - **PC0:** IP = 192.168.1.10 | Gateway = 192.168.1.1



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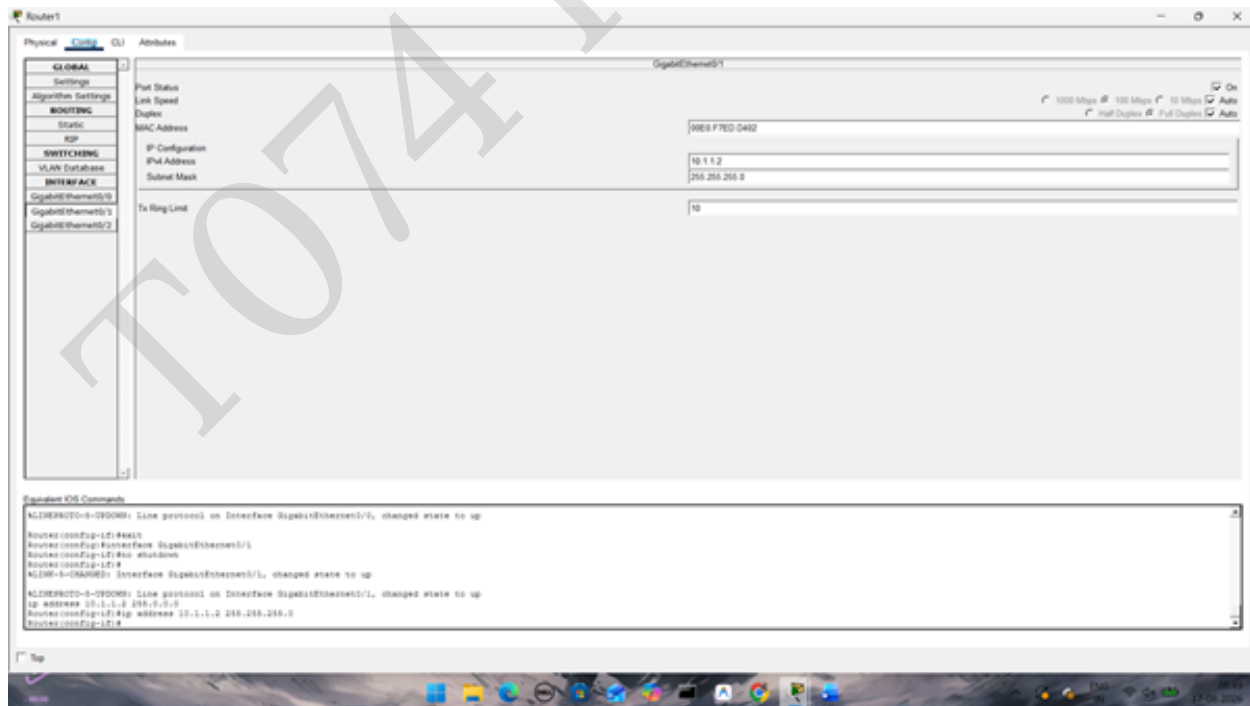
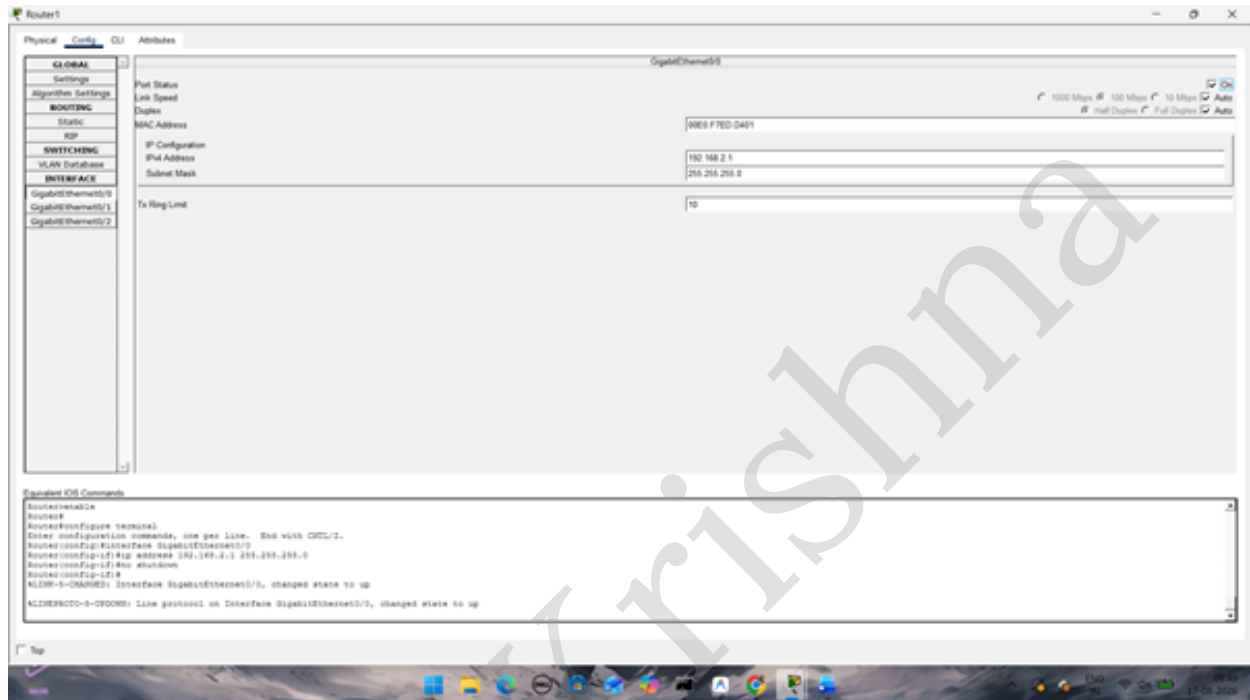


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Site 2 (Right):

- **Router1:** LAN (Gig0/0) = 192.168.2.1 | WAN (Gig0/1) = 10.1.1.2
- **PC1:** IP = 192.168.2.10 | Gateway = 192.168.2.1



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IPsec Phase 1 (ISAKMP)

- Created ISAKMP policy 10 on both routers with AES encryption, SHA hashing, and pre-shared key authentication.
- Set the shared secret key: crypto isakmp key vpnsecret123 address [Peer-WAN-IP].



IPsec Phase 2 (Transform Set & ACL)

- Created a transform set named **TS** using esp-aes and esp-sha-hmac.
- Created Extended Access Control List (ACL) **100** to permit and identify interesting traffic moving between 192.168.1.0 and 192.168.2.0.

Crypto Map Application

- Created a crypto map named **VPN-MAP** that linked the peer IP, the transform set, and the ACL.
- Applied the crypto map to the GigabitEthernet0/1 (WAN) interface on both routers to arm the VPN.



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[illegible]

Verification and Output

- **Triggering the Tunnel:** Initiated an ICMP ping from PC0 (192.168.1.10) to PC1 (192.168.2.10). The first packets timed out during negotiation, followed by successful replies, indicating the tunnel was established.
- **Verifying Encryption:** Executed the show crypto ipsec sa command on the routers. The pkts encaps (encapsulated) and pkts decaps (decapsulated) counters successfully incremented, proving the data crossing the WAN link was actively encrypted.
- **Result:** The Site-to-Site IPsec VPN was successfully deployed and verified.

